Proposal for Sidney G. Hacker Research Project

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The proposed research project is for the development of a new, non-uniform sampling (NUS) strategy in R^n with a cost-benefit analysis. Through the Spring of 2025, I will explore a variety of modern NUS strategies for continuous samples in R^n. The early stage of my research includes the exploration of various NUS strategies currently in development, the problems and fields they were developed to address, and the costs and benefits of such strategies. Following this broad, surface-level research, I will focus on one or two closely related problems and develop an inspired and creative approach to the problems and their respective NUS strategies. Finally, I will conduct informed research to develop a practical NUS strategy with a cost-benefit analysis. With the support of the Sidney G. Hacker Undergraduate Research Mathematics Scholarships, I hope to conduct this independent research under the sponsorship of Dr. Tom Asaki, knowing that this reduction in academic expenses would be of great benefit to my family and sister who will soon attend college.

Today, scientific research involves vast quantities of data point samples. The process of sample collection involves various resource costs, including material expenditure, financial costs, and the length of time taken for sampling. With the growing complexity and expense of modern technology, minimizing the resource costs between samples and the costs of sample collection is important to many fields of research, such as in nuclear magnetic resonance (NMR) spectroscopy for magnetic resonance imaging (MRI). When MRI scans are taken, there are various approaches to constructing pictures of the human body, one of which being the NMR method to study the properties and structures of atomic nuclei. To reduce the amount of radiation that affects the body, samples are taken randomly across time and the human body. This way, NUS strategies are important to optimize the health impacts of MRI scans as well as for the development of NMR spectroscopy.

In the field of optimization there is a subfield known as continuous optimization. The goal of this branch is to utilize continuous models rather than discrete models, such as in aerospace engineering. When applying NUS strategies to continuous optimization in the context of aerospace engineering, this would look like developing a model that represents an accurate resolution to describe airflow off an airplane wing. In taking various samples across notable spaces on the wing and throughout various points in time, a continuous and multidimensional mathematical model can be developed and studied. Through generalizing this mathematical model, new insights can be gained about refining NUS strategies to make them more accurate and efficient to their models. This allows NUS strategies to reduce the costs of collecting data with physical experimental research in exchange for the time required to refine mathematical understanding and practice of continuous optimization.